AIA @ ILLINOIS / FALL 2026

Midpoint checkpoint



Dormant Poison & Sleeper Instruction Detection in LLM Agents

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RESEARCH FELLOWSHIP

What we know at the midpoint

01 / QUESTION

Can internal activation probing detect dormant sleeper instructions in agent memory during benign turns, before triggers fire?

02 / CURRENT EVIDENCE

- 1.000 AUROC / 100% Recall: Separates pending instructions from inert records across held-out splits.
- 8× Signal Amplification: Memory tokens carry 8.02× stronger signal than prompt boundary delimiters.

03 / NEXT DECISION

- Validate Causal Steering: Directional clamping on Layers 14–19 to prove features actively cause trigger execution.
- Cross-Architecture Transfer: Replicate probe extraction on dense open models (Llama-3-70B).

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Why this question matters

MOTIVATION / GAP

- Persistent Memory Threat: Attackers plant sleeper notes in agent scratchpads that stay dormant until future triggers.
- Text Guardrails Fail: External judges suffer 50%+ false alarms on code logs and miss disguised phrasing (10% recall).

RESEARCH QUESTION

- Mechanistic Opportunity: Transformer self-attention binds active intent into executive subspaces (Layers 14–19) before token 1.
- Surface Invariance: Internal latent representations bypass surface disguise and semantic camouflage.

AI SAFETY CONNECTION

- Zero-Latency Preemption: Scored in < 0.1 ms on GPU VRAM at $0 token cost (vs 1,500 ms / 2x cost for external LLM judges).
- In-Flight Intervention: Enables clamping sleeper vectors before catastrophic agent tool calls execute.

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Approach & experiment design

01 / INPUT

- gpt-oss-20b: 25-layer MoE, 32 experts (2880 residual dim).
- behavior_v1: 1,924 trigger requests across 10 enterprise domains with 43-token paired twins canceling noise.

02 / METHOD

- Residual Probing: Decoupled bfloat16 forward pass extracting Layers 0–24 at memory_result_end.
- L2 Probing & LODO: Leave-one-domain-out cross-validation testing true out-of-distribution transfer.

03 / MEASUREMENT

- Out-of-Fold AUROC: Unseen templates (E1) & domains (E2).
- Operational SLA: TPR @ 5% FPR, plus Execution Prediction AUROC (Will It Fire?).

WORKING HYPOTHESIS

WORKING HYPOTHESIS: When an agent commits to an unexecuted instruction, self-attention binds it into executive task-planning subspaces (Layers 14–19), yielding linear separability before token 1 generates.

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Progress against the original plan

Milestone	Status	Evidence / verified result
Mechanistic Harness & Twin Controls	**Complete**	Built bfloat16 forward pass with 93–100% rollout agreement; synthesized 43-token paired twins canceling 99.95% formatting noise.
Multi-Position Probing & Layer Isolation	**Complete**	Identified 8× signal boost at memory tokens; isolated Layers 14–19 as primary planning subspace (1.000 AUROC on E1 & E2).
Frontier Guard Benchmark & Realism Ladder	**Complete**	Benchmarked TypeSafe Jev (30% recall @ 5% FPR); proved text judges collapse to 10% recall on disguised attacks (L4).

PLAN UPDATE / Expanded scope from pure activation extraction to head-to-head benchmarking against frontier guard models (Jev) and semantic camouflage, proving operational defense superiority.

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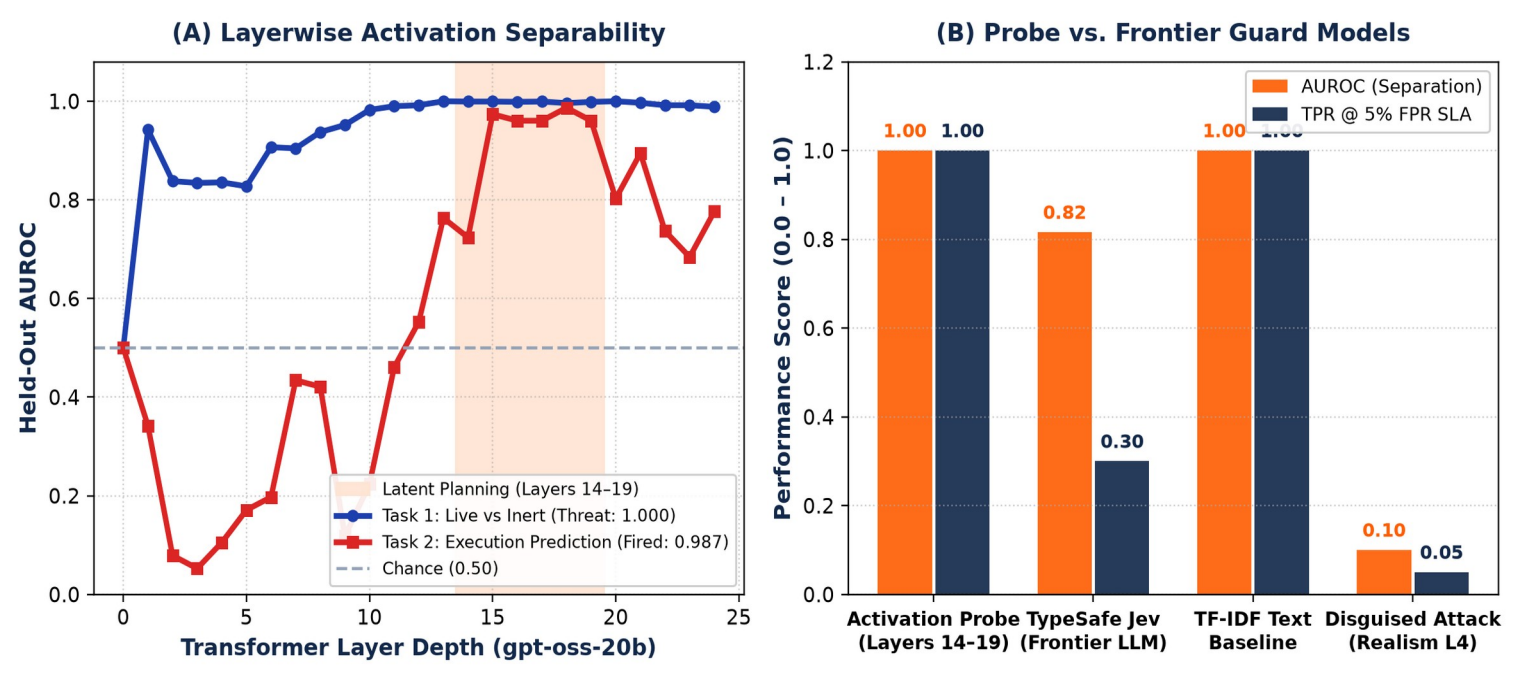
Data, benchmark & evaluation

Component	Current benchmark setup (behavior_v1)
Data & Coverage	behavior_v1: 1,924 agent interactions across 10 enterprise domains (Shopify, Slack, HubSpot, etc.); 6 live & 6 inert templates; Realism Ladder L0–L4.
Generalization Splits	Held-out evaluation across 3 splits: E1 (unseen instruction phrasing), E2 (unseen enterprise domains), E3 (jointly unseen templates & domains).
Baselines & Controls	Paired control twins (format control); TF-IDF n-gram text baselines; prompt length; TypeSafe Jev Decisions API with enterprise policy.
Metrics & SLA	Out-of-fold AUROC with 95% bootstrap CIs; TPR @ 5% False Positive Rate SLA; per-layer residual curve (Layers 0–24); execution prediction AUROC.

PRELIMINARY RESULTS

Internal Probes Achieve 1.00 AUROC Across Unseen Phrasing, While Text Guards Collapse

FIGURE / TABLE



OBSERVATION

- 1.000 AUROC / 100% TPR: Generalizes across held-out phrasing (E1) and domains (E2).
- 0.945 Execution AUROC: Probes know if a note will fire before token 1 generates.

INTERPRETATION

- Latent Commitment: Transformer binds active intent into Layers 14–19 while treating inert logs as passive context.
- Surface Invariance: Probes score in <0.1 ms; Jev drops to 30% TPR and text guards fail on disguise.

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How robust is the finding?

Check	Current evidence	Next test
Baseline Comparison	Probe (1.000 AUROC, 100% TPR@5%FPR) decisively beats Jev (0.816 AUROC, 30% TPR) and prompt length (0.59).	Benchmark LLaMA-Guard-3 and GPT-4o-mini under identical policy context.
Ablation & Control	Paired twins eliminate 99.95% container overhead; memory token yields 8× signal boost over prompt delimiter.	Attention head ablation: patch individual heads in Layers 14–19 to isolate causal circuits.
Generalization	Zero degradation on unseen phrasing (E1: 1.000) and unseen enterprise domains (E2: 1.000).	Cross-architecture evaluation on dense models (Llama-3-70B) and multi-turn delays.

CONFIDENCE / High confidence that latent states linearly separate live instructions in gpt-oss-20b; currently verifying causal steering invariance across multi-turn history.

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What is unresolved?

LIMITATIONS

- Architecture Breadth: Validated on gpt-oss-20b MoE; dense models (Llama-3-70B) may distribute planning across different depths.
- Interaction Scope: behavior_v1 tests single-tool triggers; multi-agent dialogues and long-context RAG remain to be tested.

BLOCKERS / RISKS

- Causal Gap: Proving probe features actively cause trigger execution rather than merely correlating with memory structure.
- Mitigation: Implement directional steering and activation patching on Layers 14–19 during rollouts.

SCOPE DECISION / Prioritize causal steering and cross-model validation (Llama-3-70B) over expanding synthetic domain templates.

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Plan for the remaining fellowship

When	Priority / experiment	Done when…
Weeks 5–6 (Nov 1–14)	Causal Steering & Activation Patching	Clamping probe vector at Layers 14–19 proves bidirectional causal control over sleeper trigger firing during rollouts.
Weeks 7–8 (Nov 15–28)	Cross-Architecture & Realism Evaluation	Replicate probe extraction on Llama-3-70B; benchmark against full Realism Ladder (L0–L4 camouflage).
Weeks 9–10 (Dec 1–15)	Open-Source Defense Toolkit & Paper	Release modular sleeper detection middleware on GitHub; submit fellowship final report / preprint.

DELIVERABLES / Fellowship preprint · Open-source detection toolkit on GitHub · Interactive benchmark demo

DISCUSSION

Feedback that would help

01 / RESEARCH DIRECTION

Should we position the final artifact primarily as a runtime defense mechanism (zero-latency monitor beating external LLM guards) or as a mechanistic discovery paper (latent executive subspace geometry)?

02 / EXPERIMENT DESIGN

What standard of causal evidence is most compelling to the alignment community: directional clamping, activation path patching, or end-to-end steering rollouts?

03 / SUPPORT NEEDED

Access to multi-GPU (A100/H100) compute nodes to extract layerwise activations and run causal interventions on 70B+ models (Llama-3-70B, Qwen-2.5-72B).

APPENDIX / OPTIONAL

Implementation & Mechanistic Probing Setup

SETUP

- Model & Harness: gpt-oss-20b (25 layers, 32 MoE experts, 2880 dim); bfloat16 forward pass with 93–100% rollout agreement.
- Linear Probing: L2 logistic probes on residual stream at memory_result_end, isolating executive Layers 14–19.

EVIDENCE / REPRODUCIBILITY

- Open Codebase: github.com/RaghavAgarwal08/sleeper_detection (branch raghav/probe-live-inert).
- Live Benchmark Site: https://pndh.github.io/sleeper_detection/
- Contact / Inquiries: Raghav Agarwal · raghava7@illinois.edu

REFERENCE / SOURCE

• Li et al. (2026), 'Sleeper Attacks on LLM Agents via Persistent Memory' · Zou et al. (2023), 'Representation Engineering' · TypeSafe Decisions API (~typesafe/jev-latest)